



Changes in hypopharyngeal glands of nurse bees (*Apis mellifera*) induced by pollen-containing sublethal doses of the herbicide Roundup®

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HIGHLIGHTS

- Morphological and structural changes in glandular cell mitochondria.
- Early degeneration of the rough endoplasmic reticulum as in cells of senescent glands.
- Damage to hypopharyngeal gland cells compatible with apoptosis events.
- The amount of actual jelly produced has not changed.

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ABSTRACT

Decreasing pollinator populations worldwide has generated great concern and stimulated countless studies to understand the origin of colony losses. One main cause is the indiscriminate use of different pesticides, producing subtle negative effects on bee physiology and behavior. Royal jelly synthesized in the hypopharyngeal glands is an essential protein for feeding all individuals of the hive, especially the queen. Therefore, the present study aimed to determine the effect of sublethal concentrations of Roundup® on the hypopharyngeal glands of nursing workers, including its interference with the production of royal jelly. The herbicide was found to promote changes in the cellular ultrastructure of these glands, causing early degeneration of the rough endoplasmic reticulum and morphological and structural changes in the mitochondria. No changes were noted in the amount of royal jelly produced, but additional long-term studies are necessary to determine possible qualitative changes. This is the first study to evaluate the effect of Roundup® on the royal jelly-producing glands, showing that resultant alterations in these structures can trigger damage to the development and survival of bee colonies.

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1. Introduction

In *Apis mellifera*, the hypopharyngeal glands are part of the digestive system, and they are responsible for the synthesis of

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enzymes that metabolize the nutrients released by pollen digestion and the production of royal jelly (Huang et al., 1989; Costa and Cruz-Landim, 1999). Royal jelly, a protein-based substance, is essential for the feeding of all offspring in the first three days of larval life and, for the queen, throughout her entire life (Deseyn; Billen, 2005; Feng et al., 2009; Huang et al., 2012; Kamakura, 2012).

Huang et al. (1989) verified that the development of hypopharyngeal glands in worker bees can vary significantly since they are affected by factors inside and outside the colony. The different

activities carried out by individuals of the same age and the presence of young larvae are essential for the activation of the hypopharyngeal glands (Simpson et al., 1968; Huang et al., 1989; Lass and Crailsheim, 1996; Pernal and Currie, 2000; Deseyn and Billen, 2005). As the workers begin to perform foraging activity, both Moritz and Crailsheim (1987) and Crailsheim and Stolberg (1989) found that acini of the glands and proteolytic activity decrease.

In field conditions, bees can be exposed to multiple pesticides for long periods, resulting in significant negative impacts on their colonies (Zaluski et al., 2017), particularly the hypopharyngeal glands. The action of neonicotinoid and acaricidal pesticides, which are used in crops and in the hives, reduces the size of the glandular acini owing to cell death in these structures (Alaux et al., 2010; Smodis-Skerl and Gregorc, 2010; Hatjina et al., 2013). In addition, feeding bees exposed to fipronil and pyraclostrobin, either alone or combination, within a period of six days presented alterations in the mandibular and hypopharyngeal glands (Zaluski et al., 2017).

Glyphosate-based herbicides (GBH) are the most utilized (45% of the total) pesticides in Brazil (Carneiro, 2015). Many studies have reported on the mode of action of GBH in bees. Some studies showed that 2.78–31.3 mg e./kg and 87.2–629 mg e./kg of GBH, present in nectar and pollen, respectively, do not interfere with larval survival until the seventh day (Thompson et al., 2014). In adult workers, survival is not compromised by GBH doses of 2.5–5.0 mg L⁻¹ (Herbert et al., 2014) and 1.25, 2.5 and 5 ng/bees (Helmer et al., 2015). Yet, observations made by Helmer et al. (2015) showed a decrease of β -carotene in bees submitted to GBH-contaminated feed and a small reduction in protein levels. According to Herbert et al. (2014), adult *A. mellifera* workers exposed to experimental feed contaminated with glyphosate presented reduced sucrose sensitivity, leading to loss and difficulty in establishing associative memories, which, in turn, could cause inefficient collection of nectar and pollen for the colony and, finally, compromise survival. Jumarie et al. (2017) used the thiobarbituric acid reactive substances test (TBARS) to evaluate lipid peroxidation induced by herbicides based on atrazine and glyphosate and found that mixtures of these herbicides caused a significant increase in TBARS, with levels ranging from 30 to 40% in doses 1% lower than LD50.

It is important to consider that the commercial formulations of GBH also contain the so-called “inert” ingredients that may be even more toxic than glyphosate alone (Williams et al., 2000; Mesnage et al., 2012; Sribanditmongkol et al., 2012). Some of these inert ingredients may also have toxic effects on bees, contributing to a decrease in the health of their populations (Zhu et al., 2014; Mullin et al., 2016) and impairing their learning performance (Ciarlo et al., 2012).

As herbicides are not intended to be used against insects, manufacturers do not provide any caution against bee exposure on their labeling (Johnson, 2015). As a consequence, bees can be in contact with high concentrations when that herbicide is applied in agroecosystems, particularly over genetically modified crops that are attractive to the pollinators during the flowering period such as soybeans (Pettis et al., 2013).

In the present study, we used the commercial formulation of Roundup® instead of glyphosate active ingredient, in order to simulate real agricultural conditions. In order to determine the effect of sublethal concentrations of this herbicide on the dynamics of *A. mellifera* colonies, analyses were performed on the hypopharyngeal glands of adult workers fed with pollen containing Roundup®, as well as evaluations on the production of royal jelly. Compared to bees fed with pollen without herbicide, the hypopharyngeal glands of bees exposed to Roundup® sustained both physiological and mitochondrial damage, as well as compromising the production of royal jelly. Such damage, in turn, had several

negative effects on colony dynamics.

2. Material and methods

2.1. Bee source

The hives that composed the bioassays were part of the Experimental Apiary located in Florianópolis, Santa Catarina - Brazil (27° 32'12.28 "S, 48° 30'5.82" W) established in 2012. Before the beginning of the bioassays, the hives were managed for six months to achieve homogeneity regarding the breeding area and number of adults (Palacios, 2011). The apiary is surrounded east and north by secondary forest belonging to an environmental conservation unit with an area of 491.5 ha, and west and south by two small villages. In addition, the apiary is far apart by 1.5 km at west side. Since there is not commercial agriculture other than small familiar gardens, the use of pesticides on areas up 10 km from the apiary is absent.

2.2. Herbicide

In the treatment with herbicide, it was used Roundup® Original (Concentrated Solution - COMPOSITION: Isopropylamine salt of N-(phosphonomethyl) 480 g/L, equivalent to N-(phosphonomethyl) glycine acid (GLIFOSATE) 360 g/L; The amount of Roundup® used in the bioassays was proportional to what the manufacturer recommends for application of the product in the field.

2.3. Morphology, histology and ultrastructure of hypopharyngeal glands

The bioassay was composed of two treatments: the control treatment, in which the hives received 200 ml of inverted sugar syrup (with free entrance of pollen) and “Roundup®” treatment, in which the hives received 200 mL of inverted sugar syrup mixed with 80 g of pollen and 1.5 μ L of Roundup®. This formulation offered to Roundup® treatment hives conferred a glyphosate/artificial food ratio of 2.16 μ g a.i. g⁻¹, which were administered in internal surface feeders. It should be noted that this ratio is in the same order of magnitude (10⁻³ g/kg) of the ratios found in the collection conditions of the herbicide (Thompson et al., 2014). Feeding to the bees was provided weekly for 28 days and all food was collected by the bees within 12 h and stored inside the combs.

Control hives had access to the foraging area surrounding the apiary, in addition to 200 ml of inverted sugar syrup provided weekly. Daily measurements during a week previously to the bioassay, the amount of pollen stored weekly in the collectors weighted approximately 80 g. Thus, the feed offered to hives submitted to the Roundup® treatment was composed by 80 g of pollen obtained in the collectors fixed in the carton box and 200 ml of inverted sugar syrup, to which it was added the Roundup® to final concentration of 1.5 μ L. Therefore, the single difference among both treatments is the presence or absence of residues of Roundup, since the syrup was the same and the pollen also were collected by bees in the same foraging area.

At 21 days from the beginning of artificial feeding, combs were collected to obtain the bees used in evaluations of the hypopharyngeal glands. These bees emerged in a Bio-Oxygen Demand (BOD) incubator, and the thorax was marked with nontoxic paint (Posca Paint Pens, Mitsubishi Pencil, Japan). Bees were then returned to their original hives, at which point the last artificial feeding was given for each treatment. On the sixth day after reintroduction to the hives, bees were collected for evaluation, and the pollen collectors were removed.

Five bees from each of three hives (in the total of 15 bees per

treatment) were immobilized and their heads sectioned and fixed in 5% glutaraldehyde buffered in 0.1 M Sodium cacodylate, remaining under these conditions for 24 h. Subsequently, they were dissected for extraction of the hypopharyngeal glands, which were kept in 0.1 M Sodium cacodylate buffer until the post-fixation step with 1% Osmium Tetroxide, followed by the dehydration step in increasing concentrations of alcohol and subsequent infiltration in Spurr's resin. Samples already infiltrated and included in resin were oven-dried at 70 °C for 24 h for complete polymerization of the resin, after which blocks of samples (in resin) were processed for ultramicrotomy.

Morphological evaluations were performed using Opticam stereoscopic microscope images processed with Opticam Microscopy software (OPTHD - 3.7.4103) at the Agricultural Entomology Laboratory, Agrarian Sciences Center, (Federal University of Santa Catarina - UFSC).

Histology analysis consisted of the transference of semfinished (700 µm) sections to glass slides and staining with Toluidine blue. Afterwards, they were observed in light field under optical microscopy, and the images obtained were processed with the LAS-AF Lite program (Leica). To access the ultrastructure features, ultrathin sections (60–70 nm) of the hypopharyngeal glands contrasted with 5% Uranyl Acetate and Lead Citrate were analyzed in the JEM-1011 Transmitting Electron Microscope (TEM) at an acceleration voltage of the electrons of 80 kV. The images obtained were processed with Gatan Digital Micrograph software. Samples were prepared and the images recorded at the Central Laboratory of Electronic Microscopy (LCME) of UFSC.

2.4. Production of royal jelly

The production of royal jelly was obtained in November 2017, following the method described by Doolittle (1899). The royal jelly was obtained by the production of queens, which consists of the

transfer of larvae from their original cell in the combs to artificial domes containing royal jelly. The bioassays for the production of royal jelly and larval acceptance were conducted in the same hives used in the bioassays to obtain bees for analysis of the hypopharyngeal glands. These hives were orphaned 24 h prior to introduction of the combs with larvae, the pollen collectors kept at the entrance and fed according to the corresponding treatment.

First-instar larvae were obtained from a worker's nest of a supplier hive, which was managed to be disease-free and have both healthy queen and good posture rate. The acrylic domes were fixed in transverse slats fastened in the central part of a comb. Each batten received 40 domes, totaling 80 in each comb for the production of royal jelly. The transfer of the larvae from the combs to the acrylic domes was carried out with the aid of a larvae grafting needle. Each dome contained a drop of royal jelly diluted in distilled water (1: 1).

The combs containing 80 domes with the transferred larvae were introduced into the hives that composed the treatments and evaluated after 72 h. The royal jelly collected was stored in 1.5 ml microtubes and kept under refrigeration (−20 °C). The number of accepted larvae and the total weight of jelly produced in each of the three hives of both treatments were evaluated. Data were submitted to analysis of variance, using a completely randomized design. The variables that presented significant differences at the 5% probability level ($\alpha=0.05$) by the F test were submitted to the Tukey mean separation test at the same level of significance.

3. Results and discussion

Morphological evaluations showed acini to be spherical, according to type 4 hypopharyngeal glands, as described by Meliponini and Apini (Costa et al., 2002; Cruz-Landim, 1992, 2009). However, morphological differences were observed in bee glands from hives that received Roundup®-containing food (Fig. 1-A),

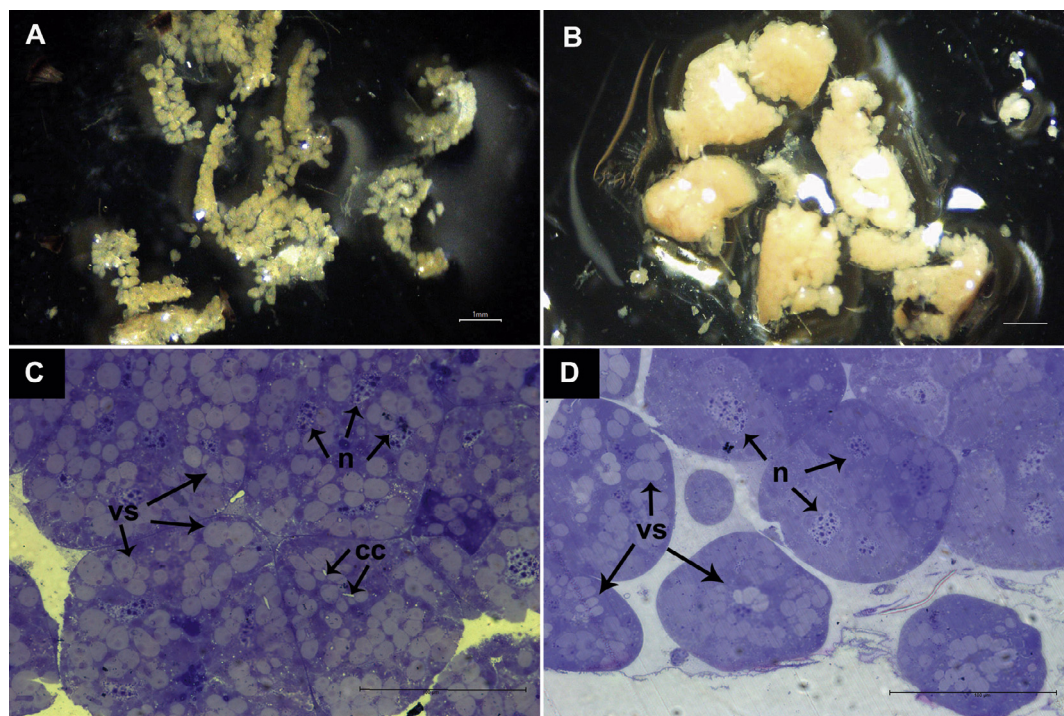


Fig. 1. Hypopharyngeal glands of bees (*A. mellifera*) at the age of six days after emergence. A – worker bee glands from control treatment hives; B – worker bee glands from a beehive that received food containing Roundup®, presenting spongy turgid acini. Images registered in magnifying glass, 1 mm scale bar. C and D – Micrographs of sections of hypopharyngeal glands stained with Toluidine blue. Individuals of the control treatment are represented by micrograph C, and micrograph D represents treatment individuals who received Roundup®-containing food. cc = collecting canaliculus; n = nucleus; vs = secretory vesicles. Scale bar 100 µm.

when compared to control individuals (Fig. 1-B). In 70% of the individuals fed with pollen containing Roundup®, the glands presented a spongy appearance with large and extremely turgid acini, suggesting the accumulation of secretion in their interior.

According to Maurizio (1954), the appearance and sizes of the acids of the hypopharyngeal glands reflects their functional status. Deseyn and Billen (2005) and Elaidy (2014) stated that the activity of protein synthesis in acini cells is asynchronous and that large glands may experience inhibition of protein synthesis if excretion of the secreted material does not occur owing to a negative feedback mechanism (Huang et al., 1989). It is possible that the GBH treatment given to hives promoted physiological changes in the bees, interfering with the functioning of the hypopharyngeal glands by inducing the accumulation of synthesized substances inside the cells.

3.1. Histology of the hypopharyngeal glands

Histology studies included evaluation and recording of the distribution of secretory cells, the size and shape of the nuclei, the presence of secretory vesicles of royal jelly, and the degree of coloration of the cellular cytoplasm. Results showed that the structure of the hypopharyngeal glands is in accordance with previous descriptions for six-day-old bees after emergence (Cruz-Landim and Hadek, 1969; Deseyn and Billen, 2005; Smodiš-Skerl and Gregorc, 2015). Histologically, they were preserved (Fig. 1C and D), with the secretory cells grouped in their interior, containing several vesicles of royal jelly, intracellular channels and collecting ducts. These characteristics are in agreement with hypopharyngeal glands described for bees of the species *A. mellifera* (Costa et al., 2002; Cruz-Landim, 1992, 2009; Suwannapong et al., 2010; Al-Ghamdi et al., 2011).

The distribution of the secretory cells within the acini and the degree of cytoplasm staining are similar between the two treatments. It was possible to clearly observe irregular size and shape of nuclei as a result of increased secretory vesicles in the cytoplasm and large number of nucleoli in their interior.

Comparing micrographs of control bees with those receiving food containing Roundup®, secretory vesicles in larger quantities were observed, indicating changes in the pattern of protein synthesis in the glands of these individuals. In studies on the effects of pesticides (insecticides, acaricides and fungicides) on the hypopharyngeal glands, the occurrence of apoptotic and necrotic cell death in these structures was described (Smodiš-Skerl and Gregorc, 2010), as well malformations and reduction in the size of acini owing to effect of pesticides (Smodiš-Skerl and Gregorc, 2010; Heylen et al., 2011; Hatjina et al., 2013; Renzi et al., 2016; Zaluski et al., 2017). While the present study did not investigate the occurrence of cell death caused by Roundup®, it will be important to perform future specific studies to evaluate this event.

3.2. Ultrastructure of hypopharyngeal glands

The cytotoxic effect of Roundup®-containing food on hypopharyngeal gland cells was performed with the use of transmission electron microscopy (TEM). Significant differences were observed in the glandular cells of Roundup®-treated bees when compared to control. These differences were more pronounced in the rough endoplasmic reticulum (RER) and mitochondria. In normal conditions, active secretory cells present well developed and organized RER, indicating intense synthesis of protein material for export, as well as vesicles of secretion of royal jelly and enzymes (Costa et al., 2002; Cruz-Landim, 1992, 2009). Mitochondria are considered reliable biosensors by which to evaluate the potential toxicity of chemical compounds *in vitro* and *in vivo* in order to understand and

predict adverse effects of numerous substances (Pereira et al., 2012).

In the micrographs of the present work, it was possible to identify the RER and vesicles of royal jelly secretion, and in the control treatment, these vesicles appeared in greater quantity (Fig. 2, A and C), indicating more intense protein synthesis activity. In hives treated with Roundup®, the RER was also found to be fragmented and discontinuous with large spaces between its membranes (Fig. 2, B and D), resembling bean cells that are no longer growing and thus show regression of the cells that make up the hypopharyngeal glands. This aspect of the RER was observed and described previously by Deseyn and Billen (2005) and Elaidy (2014), who evaluated the ultrastructure of hypopharyngeal glands in 27-day-old forage bees.

In addition, the cytoplasm of some cells in the Roundup®-containing food-treated group presented more electron density (Fig. 2-B) and/or were edematous (Fig. 2-D), forming an organelle-free cytoplasmic mass similar to that described for apoptotic cells. In hepatocytes L-02, characteristic cellular alterations of apoptosis and necrosis were observed, possibly triggered by the presence of different concentrations of Roundup® (Luo et al., 2017). Previously, GBH has been shown to stimulate the development of metabolic changes related to collapse in ATP synthesis (Gomes and Juneau, 2016; Luo et al., 2017).

Few specific studies have reported on the effects that Roundup®, other GBH, or even glyphosates have on bees. However, cellular damage observed in nontarget organisms, such as those described by Gomes and Juneau (2016) and Luo et al. (2017), is also likely to occur in bees and can cause changes in cell ultrastructure such as that observed in the present work.

The analysis of the micrographs of bee glands exposed to the feeding containing Roundup® also revealed morphological alterations in the mitochondria (Fig. 2, F), which presented, among other alterations, an annular shape, different from those exhibited by the micrographs of the control group (Fig. 2, E). According to Ding et al. (2012), mitochondria undergo dynamic structural changes to maintain homeostasis in response to mitochondrial oxidative damage in the form of a ring or in the form of a “C”.

Our results also demonstrated that the membranes of the mitochondrial ridges are more intact in the micrographs of individuals in the control group, which could be clearly identified (Fig. 2 F). According to Cruz-Landim (2009), during the secretory phase, the cells of the hypopharyngeal glands show mitochondria with slightly sharp ridges. However, the glyphosate active ingredient and the Roundup® herbicide cause changes in cell metabolism of nontarget organisms which include modifications in the mitochondrial ridges, interference in the metabolic pathway of glutathione synthesis with an imbalance of the oxidation and antioxidant system, alterations in the electron transport chain with release of cytochrome c and mitochondrial cytoplasm, significant increase of reactive oxygen species (ROS), reduction of energetic functions of mitochondria, lipid peroxidation and increase in the permeability of cell membranes (Peixoto, 2005; Gomes and Juneau, 2016; Jumarie et al., 2017; Luo et al., 2017).

No morphological changes were found in the cell nucleus structure in the analyzed micrographs, irrespective of treatment (Fig. 2G and H). This could be explained by the short time that bees were exposed to the feed containing Roundup®. It is also possible that the herbicide did not promote changes in these structures. Nevertheless, the damage observed in mitochondria and granular endoplasmic reticulum is sufficient to demonstrate how harmful this herbicide is to the cells of the hypopharyngeal glands and how much this can compromise the viability of the colonies, considering the great importance of these structures in the production of royal jelly.

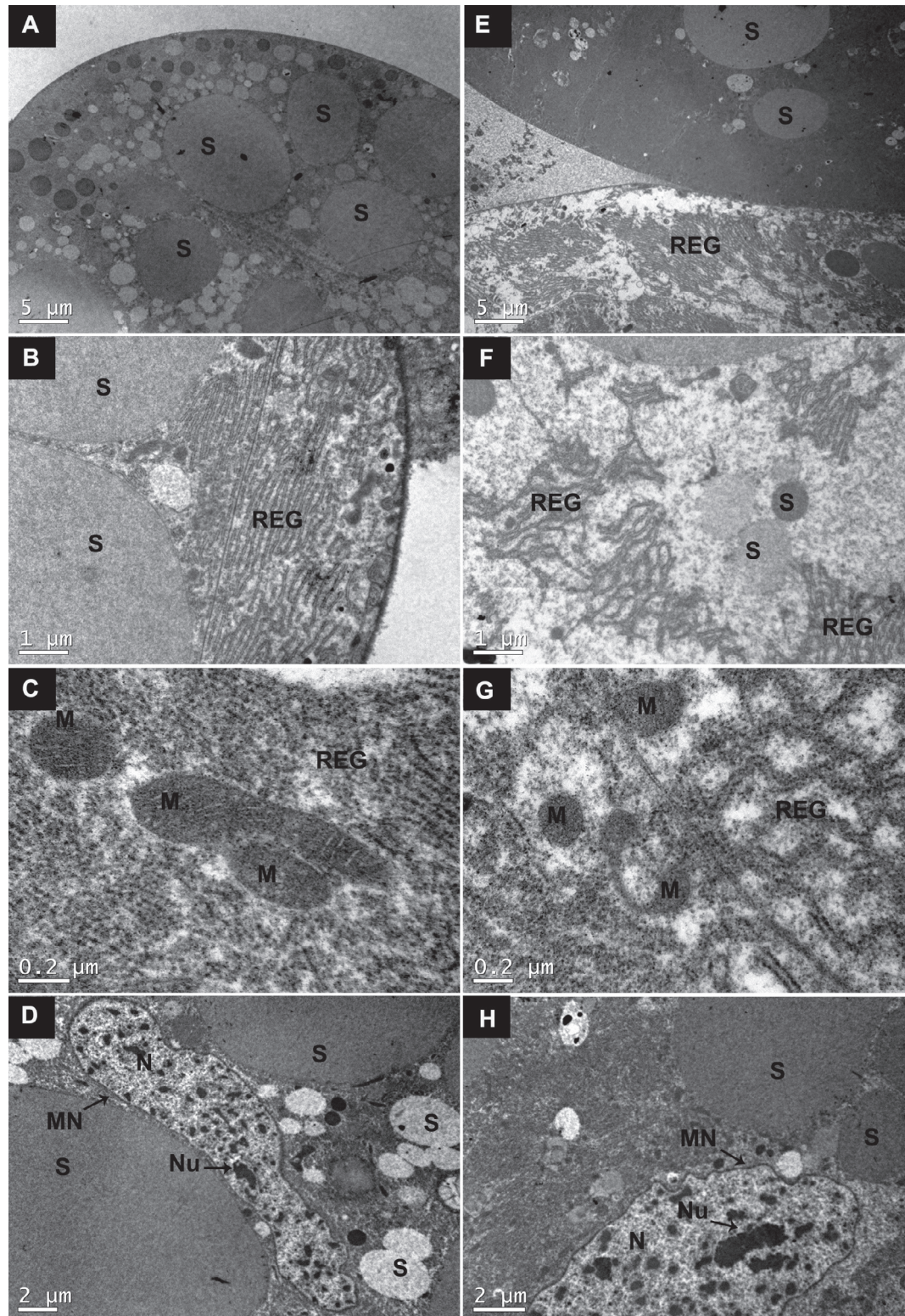


Fig. 2. Transmission electron micrograph (TEM) of hypopharyngeal gland cells of six-day-old *A. mellifera* workers after emergence. Treatment control: A, C, E and G; treatment with Roundup®: B, D, F and H. The A-D micrographs represent the cytoplasm of the cells that make up the glandular acini. It is possible to observe regions fragmented in the RER in micrographs of the treated group (B and D) and intact in the control group (A and C). Micrographs E and F present the mitochondria, in which it is possible to observe organelles with annular shape in the cells of the treated group (F). In micrographs G and H, it is possible to observe the nucleus with multiple nucleoli and complete library. Secretion (S); mitochondria (M); rough endoplasmic reticulum (RER); nucleus (N); nucleolus (Nu).

The results obtained in the present work do not demonstrate alterations in the histology of the hypopharyngeal glands of the bees that received food containing Roundup®. However, it was possible to clearly identify alterations in the cellular ultrastructure of these individuals in which the RER was fragmented and discontinuous, indicating degeneration of these structures, as happens in the cells of the bees. The premature degeneration of these cells may cause worker bees to anticipate the foraging task, decreasing the number of individuals remaining in the hive performing care tasks with the offspring. One of the consequences would be the reduction in litter production, with negative effects on the development and maintenance of the colony and also a reduction in the lifespan of adult workers (Hatjina et al., 2013).

Damage to the cells of the hypopharyngeal glands can also affect the quantity or quality of the actual jelly produced, compromising the feeding of the queen and the larvae in their early stages of development. Studies conducted by Wessler et al. (2016) have shown that bees subjected to chronic neonicotinoid exposure have shown an 80% reduction in non-neural acetylcholine (ACh) synthesis owing to the death of cuticular cells of the collector channels where ACh is produced. Although the biological role of ACh is not fully known for the larval development of bees, the authors consider the possibility that ACh is involved with their trophic development, as well as representing an ammonium compound with antimicrobial action in these larvae (Wessler et al., 2016).

3.3. Production of royal jelly

Evaluations to verify possible changes in the production of royal jelly, as a result of the treatments applied to the hives, revealed no statistically significant differences in the acceptance of larvae ($F_{1,4} = 0.053$; $p = 0.82$) or the average weight of royal jelly by queen cell ($F_{1,4} = 2778$; $p = 0.17$), at 5% probability. Nevertheless, the hives that received food containing Roundup® presented an average weight of royal jelly lower than that of control hives (0.167 g and 0.287 g per queen cell, respectively). According to Li and Aiping (2003) and Garcia and Couto (2005), biological and behavioral interactions of the bees are related to the production of royal jelly in addition to genetic components that experience interference from factors, both internal and external to the colonies.

From the damage verified in the ultrastructure of the cells that make up the hypopharyngeal glands, as described in this work, it is possible to infer that the time of exposure to sublethal doses of the herbicide was not sufficient to reduce the amount of royal jelly produced by the colonies at a statistically significant level. However, in addition to the quantification, it would be important to analyze the nutritional quality of the actual royal jelly produced and to verify if any changes had occurred, depending on the treatments.

4. Conclusions

To the best of our knowledge, this is the first study to evaluate the effects of Roundup® herbicide on the hypopharyngeal glands of nursing bees and the consequences of the production of royal jelly.

Based on our results it is possible to affirm that the feed containing Roundup® supplied to the bees caused changes in the cellular ultrastructure of the hypopharyngeal glands, which lead to an early degeneration of RER compatible with the aging, and regression of the hypopharyngeal glands. Thus, glandular cells of six-day-old nourishing bees exhibited the appearance of cells of forage bees more than 25 days after emergence. Morphological and structural changes were also verified in mitochondria that could decrease its bioenergetic functions, and trigger other cellular damage related to ATP depletion, such as apoptosis. However,

additional studies, such as the involvement of oxidative enzymes (e.g.: glutathione peroxidase), levels of ROS and lipid peroxidation, are needed to deeply understand the causal mechanisms. No significant reduction in larval acceptance in queen-rearing artificial domes or in the production of royal jelly was observed, but because of cellular alterations described here, it is necessary to carry out specific evaluations on the quality of this product.

Taking into account the importance of royal jelly both for larvae and for all adult individuals in the hive (Crailsheim, 1992, 1998), it is reasonable to presume that the scientific advancement revealed in the present study supports that sublethal doses of Roundup® produce subtle negative effects on bees that may compromise the long-term survival of their colonies. However, the results presented here are not representative of actual field conditions to which these insects are exposed, because simultaneously a large amount of distinct pesticides are used frequently, which are then transported and stored inside the hives, exerting synergistic and prolonged effects on hive dynamics (Goulson et al., 2015), and possibly potentiating damage to the hypopharyngeal glands and mitochondria, as well as interfering with the quality of the royal jelly.

Considering the importance of pollinators for the maintenance of the balance of different ecosystems (Kremen, 2004; Eardley et al., 2006; Imperatriz-Fonseca and Nunes-Silva, 2010; Mouga and Mouga, 2012; Toledo et al., 2013), as well as for the production of food in agroecosystems and bee products, it is urgent that regulatory agencies and public policies take into account the evidence-based adverse effects on bees that have been obtained by the independent scientific community. These alarming facts also call for measures and efforts to minimize bee loss, regardless of species, in different locations around the world.

Declarations of interest

None.

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